

Path-integral bosonization in topologically nontrivial sectors: The non-Abelian case

D. Cabra and M. V. Manias

*Departamento de Física, Universidad Nacional de La Plata, CC 67, 1900 La Plata, Argentina
and Consejo Nacional de Investigaciones Científicas y Técnicas, Argentina*

F. A. Schaposnik

*Laboratoire de Physique Théorique Particules Élémentaires, Université de Paris 7,
Tour 24-5 et., 2 Place Jussieu, 75251 Paris CEDEX 05, France,
Departamento de Física, Universidad Nacional de La Plata, CC67 1900 La Plata, Argentina,
and Comisión de Investigaciones Científicas Buenos Aires (CICBA), Buenos Aires, Argentina*

M. Trobo

*Departamento de Física, Universidad Nacional de La Plata, CC 67, 1900 La Plata, Argentina
and Consejo Nacional de Investigaciones Científicas y Técnicas, Argentina
(Received 12 December 1990)*

We present a path-integral bosonization approach which allows us to take into account nontrivial topological sectors in two-dimensional models such as two-dimensional QCD. Because of the existence of fermionic zero modes, the partition function receives a contribution only from the topologically trivial ($n=0$) sector. However, certain fermionic correlation functions (constructed from chiral-charge-nonconserving operators) become nonvanishing when one takes into account $n \neq 0$ sectors.

I. INTRODUCTION

The path-integral framework has proven to be very useful in the study of two-dimensional fermionic models such as the Schwinger model¹ (SM), or its non-Abelian extension [two-dimensional quantum chromodynamics^{2,3} (QCD₂)]. Indeed, the fermionic generating functional can be very simply calculated by means of a chiral change of variables, the path-integral version of the operational bosonization rules.^{4,5} The same approach has also led to new insight into purely fermionic models such as the Thirring model, Gross-Neveu model, etc.⁶⁻⁸

As we stated above, the key of path-integral bosonization is a chiral change of the fermionic integration variables. For example, for a two-dimensional gauge theory (with gauge group G) working in a certain gauge [the Lorentz gauge for $G=U(1)$,¹ the decoupling gauge for $G=SU(N)$ (Refs. 2 and 9)], this change reads

$$\begin{aligned}\psi &\rightarrow \chi = e^{\gamma_5 \phi(x)} \psi, \\ \bar{\psi} &\rightarrow \bar{\chi} = \bar{\psi} e^{\gamma_5 \phi(x)},\end{aligned}\quad (1.1)$$

with $\phi(x)$ taking values in the Lie algebra of G .

Transformations (1.1) are intended to decouple the gauge field (or auxiliary field, in the case of purely fermionic models) from fermions. Indeed, under transformations (1.1), the fermionic original Lagrangian L_F ,

$$L_F = \bar{\psi}(i\partial + A)\psi = \bar{\psi}\mathcal{D}[A]\psi, \quad (1.2)$$

becomes

$$L_F = \bar{\chi}i\partial\chi \quad (1.3)$$

provided one has chosen a gauge condition such that the gauge field can be written in the form

$$A = -e^{\gamma_5 \phi(x)} i\partial e^{\gamma_5 \phi(x)}. \quad (1.4)$$

Condition (1.4) corresponds to the Lorentz gauge for $G=U(1)$ and to the so-called decoupling gauge for $G=SU(N)$ (Refs. 9 and 8) but the decoupling can be performed in an arbitrary gauge.¹⁰

Of course, this is not the whole story. There is, at the quantum level, a “Fujikawa Jacobian”¹¹ associated with (1.1). In order to define this Jacobian, let us write the fermionic measure in the generating functional in the form

$$D\mu_A[\psi] = D\bar{\psi} D\psi e^{-S_F[\bar{\psi}, \psi, A]}, \quad (1.5)$$

which explicitly shows its dependence on the gauge field A_μ . Then, we define the Jacobian $J[A]$ through the equation

$$D\mu_A[\psi] = J[A] D\mu_{A=0}[\chi]. \quad (1.6)$$

One then gets, after the integration of (1.6),

$$J[A] = \frac{\det \mathcal{D}[A]}{\det i\partial}. \quad (1.7)$$

When one deals with gauge theories, a gauge-invariant regularization prescription (such as ζ -function method) has to be employed. Once this is done, one discovers that it is the noninvariance of the fermionic measure under transformations (1.1) which is responsible, in the path-integral framework, for the Schwinger mechanism.¹² Indeed, a mass for the gauge field A_μ is generated through the Jacobian (1.7) in the Abelian case¹ [for

$G = \text{SU}(N)$, a Wess-Zumino¹³ term is generated^{2,3}].

We thus see that transformations (1.1) play the same role as the bosonization rules^{4,5} in the operational approach since, after performing the change of variables one is left with a bosonic Lagrangian which corresponds to the one gotten by usual bosonization (see Ref. 14 for details).

There is, however, a loophole in the scenario above. Namely, in going from Lagrangian (1.2) to Lagrangian (1.3) one is passing from a Dirac operator $\mathcal{D}[A]$ defined on some topological sector n (n labels the topological charge of the gauge field A_μ) to $i\partial$, i.e., to a Dirac operator with $A_\mu = 0$, i.e., defined in the $n=0$ sector. Thus, transformations (1.1) are in general topology changing, this implying that one cannot naively apply the usual regularization prescriptions when computing J . Indeed, these prescriptions (ζ function, heat kernel, etc.) are well defined only for compact manifolds,¹⁵ but if one allows topology-changing transformations, the space-time manifold cannot be, in principle, compactified.

It is interesting to note at this stage that the partition functions Z gotten in Refs. 1–3, 6, and 7 for different models, ignoring the problem signaled above, agree with those arising from the operational approach. This may indicate that only the zeroth topological sector contributes to Z and that $n \neq 0$ sectors are, in principle, unimportant. As we shall see, although Z is not modified when one includes the effect of topological sectors, certain correlation functions become nontrivial only when $n \neq 0$ sectors are taken into account.

Once the problem of handling transformations (1.1) for A_μ 's carrying topological charge is recognized, one has at least two alternatives to face it. Either one develops a regularization scheme valid for transformations changing from one topological sector to another or, instead, one overcomes the problem by restricting the change of fermionic variables to a definite topological sector.

Following an idea developed in Ref. 16, this last approach has been undertaken in Refs. 17 and 18 by studying some Abelian models. It is the purpose of the present work to go further in this direction, extending the analysis to the non-Abelian case.

Now, what do we mean by saying that one should restrict the change in the fermionic variables to a definite topological sector? Consider the partition function Z ,

$$Z = \int D A_\mu e^{-S_G[A]} D\mu_A[\psi] \quad (1.8)$$

with

$$S_G[A] = \frac{1}{4e^2} \int d^2x F_{\mu\nu}^2. \quad (1.9)$$

One can make explicit the contributions to Z coming from different topological sectors by writing

$$Z = \sum_n \int D A_\mu^{(n)} e^{-S_G[A^{(n)}]} D\mu_{A^{(n)}}[\psi], \quad (1.10)$$

where the sum runs over different topological sectors n . Now, in each sector we write, instead of (1.4),

$$A = A_{\text{cl}}^{(n)} - a \quad (1.11)$$

with

$$\not{a} = -e^{\gamma_5 \phi(x)} i \not{\partial} e^{\gamma_5 \phi(x)}. \quad (1.12)$$

The topological charge is carried by the *fixed-background* configuration $A_{\text{cl}}^{(n)}$ while a_μ lives in the $n=0$ topological sector. We then have

$$Z = \sum_n \int D a_\mu e^{-S_G[A_{\text{cl}}^{(n)}]} D\mu_{(A_{\text{cl}}^{(n)} - a)}[\psi]. \quad (1.13)$$

There is now no harm in decoupling fermions from a_μ since, in doing so, one remains in the same topological sector (remember a_μ carries zero topological charge). After a change of variables such as in (1.1), the partition function reads

$$Z = \sum_n \int D a_\mu J[A_{\text{cl}}^{(n)}, a] e^{-S_G[A_{\text{cl}}^{(n)}, a]} D\mu_{A_{\text{cl}}^{(n)}}[\chi] \quad (1.14)$$

or

$$Z = \sum_n \int D a_\mu J[A_{\text{cl}}^{(n)}, a] e^{-S_G[A_{\text{cl}}^{(n)}, a]} \det \mathcal{D}[A_{\text{cl}}^{(n)}]. \quad (1.15)$$

Fermions are now not completely decoupled or, in other terms, one still has to compute a fermion determinant. The advantage of expressions (1.15) lies in the fact that the fermionic determinant which is left corresponds to a fixed background which can then be chosen so as to simplify calculations. Moreover, without any further computation one can understand from (1.15) why the generating functional computed in Refs. 1–3, 6, and 7 without taking care of topology gave the correct answer for Z . Indeed, since the Dirac operator $\mathcal{D}[A_{\text{cl}}^{(n)}]$ has n zero modes,¹⁹ only the $n=0$ term in (1.15) gives a contribution to Z . On the other hand, correlation functions having an adequate number of fermionic fields can make other topological sectors contribute, provided they cancel the contributions coming from zero modes, through Grassmann integration, as we shall explain below.

As we stated above, it is the purpose of this paper to discuss these issues in connection with a non-Abelian two-dimensional model, namely, QCD_2 (but we shall explain how our analysis can be easily extended to purely fermionic models such as the chiral Gross-Neveu model^{6,7}). The paper is organized as follows. In Sec. II we discuss the path-integral approach to bosonization based on the decoupling change of variables (1.1). Following Refs. 17 and 18 we then briefly describe how nontrivial topological sectors can be handled within this approach by discussing a simple Abelian model. We extend in Sec. III the discussion to the case of an $\text{SU}(N)$ gauge theory showing how the partition function and correlation functions are affected when topological sectors are taken into account. Finally, in Sec. IV we summarize our results and conclusions, also making some comments on other non-Abelian two-dimensional models as well as on possible applications of our approach to the study of $B+L$ violation.

II. PATH-INTEGRAL BOSONIZATION: THE ABELIAN CASE

A. The partition function

In order to exemplify how topological effects can be handled within the path-integral approach to bosonization, we briefly describe in this section the treatment given in Ref. 18 in order to solve the SM.

We work in two-dimensional Euclidean space-time and we use Hermitian γ matrices satisfying

$$[\gamma_\mu, \gamma_\nu] = 2\delta_{\mu\nu}, \quad (2.1)$$

$$\gamma_\mu \gamma_5 = i\epsilon_{\mu\nu} \gamma_\nu, \quad (2.2)$$

with

$$\gamma_5 = i\gamma_0\gamma_1, \quad (2.3)$$

$$\epsilon_{01} = -\epsilon_{10} = 1. \quad (2.4)$$

As we stated in the Introduction, we shall start from the partition function (1.10),

$$Z = \sum_n \int D A_\mu^{(n)} e^{-S_G[A^{(n)}]} D\mu_{A^{(n)}}[\psi], \quad (2.5)$$

and consider a change of A_μ variables in the form

$$A = A_{\text{cl}}^{(n)} - a \quad (2.6)$$

with $A_{\text{cl}}^{(n)}$ a fixed background configuration in the n th topological sector:

$$\oint dx^\mu A_{\mu\text{cl}}^{(n)} = -2\pi n. \quad (2.7)$$

The $A_{\mu\text{cl}}^{(n)}$ configuration can be taken as a Nielsen-Olesen vortex solution:²⁰

$$A_{\mu\text{cl}}^{(n)} = A(|x|) \epsilon_{\mu\nu} \frac{x^\nu}{|x|} \quad (2.8)$$

with $A(0)=0$, $A(\infty)=-n$. [One knows the exact form for $A(|x|)$ in a particular case, see Ref. 21.]

Working in the Landau gauge (i.e., $\partial_\mu A^\mu=0$), one can write a_μ in the form (1.12). Indeed, in the present Abelian case, Eq. (1.12) takes the simpler form

$$a_\mu = -\epsilon_{\mu\nu} \partial_\nu \phi. \quad (2.9)$$

It is then easy to see that the fermionic change of variables (1.1),

$$\begin{aligned} \psi &\rightarrow \chi = e^{\gamma_5 \phi(x)} \psi, \\ \bar{\psi} &\rightarrow \bar{\chi} = \bar{\psi} e^{\gamma_5 \phi(x)}, \end{aligned} \quad (2.10)$$

eliminates a_μ from the fermionic Lagrangian:

$$L_F = \bar{\psi} \mathcal{D}[A] \psi = \bar{\chi} \mathcal{D}[A_{\text{cl}}^{(n)}] \chi. \quad (2.11)$$

As we mentioned, there is a nontrivial Jacobian associated with the change of variables (2.10) in the path integral defining Z :

$$Z = \sum_n \int D a_\mu J[A_{\text{cl}}^{(n)}, a] e^{-S_G[A_{\text{cl}}^{(n)}, a]} D\mu_{A_{\text{cl}}^{(n)}}[\chi]. \quad (2.12)$$

The evaluation of this Jacobian using Fujikawa's technique¹¹ is, by now, standard (see, for example, Ref. 8 for a detailed analysis of Jacobians associated to chiral transformations both in Abelian and non-Abelian models). Here we just give the answer, which in the present Abelian case takes the form

$$\begin{aligned} J[A_{\text{cl}}^{(n)}, a[\phi]] &= \exp \left[\frac{1}{\pi} \int d^2x \epsilon_{\mu\nu} \partial_\mu A_{\text{cl}\nu}^{(n)} \phi \right. \\ &\quad \left. - \frac{1}{2\pi} \int d^2x (\partial_\mu \phi)^2 \right]. \end{aligned} \quad (2.13)$$

We then have, for Z ,¹⁸

$$\begin{aligned} Z &= \sum_n \exp \left[-\frac{1}{4e^2} \int d^2x (F_{\text{cl}}^{\mu\nu})^2 \right] \\ &\quad \times \int D\bar{\chi} D\chi D\phi \exp \left[-\int d^2x L_{\text{eff}}^{(n)} \right], \end{aligned} \quad (2.14)$$

with

$$\begin{aligned} L_{\text{eff}}^{(n)} &= \bar{\chi} \mathcal{D}[A_{\text{cl}}^{(n)}] \chi + \frac{1}{2e^2} \phi \left[\nabla \nabla - \frac{e^2}{\pi} \nabla \right] \phi \\ &\quad + \frac{1}{e^2} F_{\text{cl}}^{\mu\nu} \epsilon_{\mu\alpha} \partial_\nu \partial^\alpha \phi - \frac{1}{\pi} \phi \epsilon^{\mu\nu} \partial_\mu A_{\text{cl}\nu}^{(n)}. \end{aligned} \quad (2.15)$$

One then has for Z , after integrating out fermions,

$$Z = \sum_n \det(\mathcal{D}[A_{\text{cl}}^{(n)}]) Z_\phi^{(n)} \exp \left[-\frac{1}{4e^2} \int d^2x (F_{\text{cl}}^{\mu\nu})^2 \right] \quad (2.16)$$

with

$$\begin{aligned} Z_\phi^{(n)} &= \int D\phi \exp \left\{ -\int d^2x \left[\frac{1}{2e^2} \phi \left[\nabla \nabla - \frac{e^2}{\pi} \nabla \right] \phi \right. \right. \\ &\quad \left. \left. + \frac{1}{e^2} F_{\text{cl}}^{(n)\mu\nu} \epsilon_{\mu\alpha} \partial_\nu \partial^\alpha \phi \right. \right. \\ &\quad \left. \left. - \frac{1}{\pi} \phi \epsilon^{\mu\nu} \partial_\mu A_{\text{cl}\nu}^{(n)} \right] \right\}. \end{aligned} \quad (2.17)$$

[Note that we have absorbed the (trivial) Jacobian arising when one changes from a_μ to ϕ variables.]

Now, we know that the index theorem guarantees the existence of $|n|$ zero modes associated with the operator $\mathcal{D}[A_{\text{cl}}^{(n)}]$. [We consider compactified space-time; otherwise, only $(|n|-1)$ zero modes are square integrable.] For $n > 0$ ($n < 0$) they correspond to right-handed (left-handed) solutions η_R (η_L) which, in terms of light-cone variables $z = x_0 + ix_1$ and $\bar{z} = x_0 - ix_1$, take the form¹⁹

$$\eta_R^m = \begin{pmatrix} z^m h(z, \bar{z}) \\ 0 \end{pmatrix}, \quad (2.18)$$

$$\eta_L^m = \begin{pmatrix} 0 \\ \bar{z}^m h^{-1}(z, \bar{z}) \end{pmatrix}, \quad (2.19)$$

where $m = 0, 1, \dots, |n|-1$,

$$h(z, \bar{z}) = e^{f(z, \bar{z})} \quad (2.20)$$

and f is defined in terms of the Nielsen-Olesen vortex solution²⁰ (2.8) written in the form

$$A_{cl\mu}^{(n)} = -\epsilon_{\mu\nu} \partial_\nu f. \quad (2.21)$$

The asymptotic behavior of f is

$$\lim_{|x| \rightarrow \infty} f(x) \sim -n \ln|x|. \quad (2.22)$$

Of course, the fermion determinant in Eq. (2.16) has to be regularized. We know, however, that any sensible regularization should lead to a vanishing result for $n \neq 0$, due to the presence of the zero modes. We thus see that only the $n=0$ sector contributes to Z in (2.16) and, hence, at the partition function level, one does not find any new result in comparison with those obtained in Ref. 1 where topologically nontrivial sectors were, in fact, ignored.

B. Nontrivial correlation functions

We shall now see how certain correlation functions, which vanish in the $n=0$ sector, become nontrivial if one appropriately takes into account topological effects. Indeed, in order to get nonvanishing contributions from these sectors, one has to look for correlation functions in which the number of Fermi fields is such that the zero-mode contribution gets absorbed. In other terms, consider the Bessel-Fourier expansion of Fermi fields in (2.12):

$$\chi_{R,L} = \sum_i e_{R,L}^i \chi_{R,L}^i + \sum_m^{|n|-1} c_{R,L}^m \eta_{R,L}^m, \quad (2.23)$$

where $\chi_{R,L}^i$ are eigenfunctions of $\mathcal{D}[A_{cl}^{(n)}]$ with nonvanishing eigenvalues, $e_{R,L}^i$ and $c_{R,L}^m$ are Grassmann coefficients. (An analogous expansion is used for $\bar{\chi}_{R,L}$.)

From the Grassmann integration point of view, the fermionic determinant vanishes because there are $dc_{R,L}^m$ integrations not accompanied by the corresponding $c_{R,L}^m$ coefficients (and the same for the $\bar{c}_{R,L}^m$ variables associated to the $\bar{\chi}$ fields). A nonvanishing correlation function will then have a product of an appropriate number of χ and $\bar{\chi}$ fields which will provide the lacking $c_{R,L}^m$ and $\bar{c}_{R,L}^m$ coefficients in a definite sector. One can easily see that chirality-changing correlation functions of the form¹⁸

$$\left\langle \prod_{i=0}^{k-1} s_\pm(x_i) \right\rangle = F_\pm(x_0, x_1, \dots, x_{k-1}), \quad (2.24)$$

with

$$s_+(x) = \bar{\psi}_R(x) \psi_R(x), \quad (2.25)$$

$$s_-(x) = \bar{\psi}_L(x) \psi_L(x), \quad (2.26)$$

pick such a contribution from the k th topological sector. For example, one gets, for F_+ ,¹⁸

$$F_+(x_0, x_1, \dots, x_{k-1}) = \exp \left[-2 \sum_{i,j} \left[\frac{1}{2} K_0(z_{ij}) + \ln(cz_{ij}) \right] + \sum_{i \neq j} 2 \ln z_{ij} \right], \quad (2.27)$$

where $K_0(z)$ is the zeroth-order modified Bessel function and

$$z_{ij} = \frac{e}{\sqrt{\pi}} |z_i - z_j|. \quad (2.28)$$

An analogous result holds for F_- correlation functions. On the contrary, chirality-conserving correlation functions, constructed, for example, from fermionic currents $j_\mu(x)$, are not affected by zero modes. In other words, only the $n=0$ sector contributes to these functions.

III. QCD₂: THE PARTITION FUNCTION

We shall now extend the approach described in Sec. II A so as to study two-dimensional non-Abelian gauge theories with massless fermions (QCD₂). We consider gauge field $A_\mu = A_\mu^a t^a$ taking values in the Lie algebra of $SU(N)$ with generators t^a satisfying

$$[t^a, t^b] = i \epsilon^{abc} t^c, \quad (3.1)$$

$$\text{tr}(t^a t^b) = \frac{\delta^{ab}}{2}. \quad (3.2)$$

Fermions are taken in the fundamental representation and the action for the model reads

$$S[A, \bar{\psi}, \psi] = S_G[A] + S_F[A, \bar{\psi}, \psi] \quad (3.3)$$

with

$$S_G[A] = \frac{1}{2} \text{tr} \int d^2x F_{\mu\nu}^2 \quad (3.4)$$

and

$$S_F[A, \bar{\psi}, \psi] = \int d^2x \bar{\psi} (i \not{\partial} + e A) \psi = \int d^2x \bar{\psi} \mathcal{D}[A] \psi. \quad (3.5)$$

As in the Abelian case, two-dimensional gauge field configurations $A_{\mu cl}^{(m)}$ carrying a topological charge $m \in \mathbb{Z}_N$ can be found for the $SU(N)$ case. [As explained in Ref. 22, the relevant homotopy group is in this case $\mathbb{Z}_N$ and not $\mathbb{Z}$ as in the $U(1)$ case.]

If we call φ the angle characterizing the direction at infinity, a mapping $g_m(\varphi) \in SU(N)$ belonging to the m th homotopy class ($m=0, 1, \dots, N-1$) satisfies, when one turns around a close contour

$$g_m(2\pi) = \exp \left[\frac{2\pi i m}{N} \right] g_m(0). \quad (3.6)$$

This can be achieved just by taking g_m in the Cartan subgroup of the gauge group. For example, in the $SU(2)$ case, one can take²²

$$g_m = \exp[i t^3 \Omega_m(\varphi)] \quad (3.7)$$

with

$$\Omega_m(2\pi) - \Omega_m(0) = (2k + m)2\pi, \quad (3.8)$$

where $m=0, 1$ labels the topological charge and $k \in \mathbb{Z}$. More general configurations in a given topological sector can be obtained by gauge rotations.

We can now write a gauge field configuration belonging

to the m th topological sector:

$$A_{\mu\text{cl}}^{(m)} = -i A(\rho) g_m^{-1} \partial_\mu g_m, \quad (3.9)$$

with g_m given by Eq. (3.7) [or its extension to the $SU(N)$ case] and

$$A(0) = 0, \quad A(\infty) = \frac{1}{e}. \quad (3.10)$$

These and more general configurations have been studied in detail in Refs. 22 and 23.

We then consider a gauge field configuration as in Eq. (3.9) as a background writing

$$A = A_{\text{cl}}^{(m)} - a \quad (3.11)$$

with $A_{\mu\text{cl}}^{(m)}$ such that

$$A_{+\text{cl}}^{(m)} = 0. \quad (3.12)$$

It will be convenient to work in a light-cone-like gauge,²⁴

$$A_- = A_{-\text{cl}}^{(m)}, \quad (3.13)$$

and write A_+ in the form

$$A_+ = -ih^{-1} \partial_+ h \equiv a[h] \quad (3.14)$$

with $h \in SU(N)$.

With this, the gauge field measure DA appearing in the QCD_2 partition function

$$DA = \Delta_{\text{FP}} \delta(A_- - A_{-\text{cl}}^{(m)}) DA_+ DA_- \quad (3.15)$$

[here Δ_{FP} is the Faddeev-Popov determinant associated with gauge condition (3.13)] takes the form

$$DA = \Delta_{\text{FP}} \left| \frac{\delta A_+}{\delta h} \right| Dh. \quad (3.16)$$

Now, the Faddeev-Popov determinant can be easily seen to be

$$\Delta_{\text{FP}} = \det(D_- [A]^{\text{adj}}), \quad (3.17)$$

where the superscript adj indicates that the covariant derivative should be taken in the adjoint representation. Concerning $|\delta A_+ / \delta h|$, one can see from Eq. (3.14) that it takes the form

$$\left| \frac{\delta A_+}{\delta h} \right| = \det(D_+ [A]^{\text{adj}}). \quad (3.18)$$

Using Eqs. (3.17) and (3.18) we have, for DA ,

$$DA = \det(D [A]^{\text{adj}}) Dh. \quad (3.19)$$

Now, one can relate the determinant in Eq. (3.15) with that corresponding to D in the fundamental representation. The answer is¹⁰

$$\det(D [A]^{\text{adj}}) = \det^{2N} D [A]. \quad (3.20)$$

After the change (3.11) we have, for the QCD_2 partition function Z ,

$$Z = \sum_m \int DA^{(m)} D\bar{\psi} D\psi \exp(-S), \quad (3.21)$$

the expression

$$\begin{aligned} Z = \sum_m \int Dh D\bar{\psi} D\psi \det^{2N} D [A_{\text{cl}}^{(m)} - a[h]] \\ \times \exp\{-S_G[A_{\text{cl}}^{(m)} - a[h]] \\ - S_F(A_{\text{cl}}^{(m)} - a[h], \bar{\psi}, \psi)\}, \end{aligned} \quad (3.22)$$

or

$$Z = \sum_m \int Dh \exp(-S_A[A_{\text{cl}}^{(m)} - a[h]]) \det^{2N+1} D [A_{\text{cl}}^{(m)} - a[h]]. \quad (3.23)$$

In order to get the effective bosonic action for QCD_2 , we just now have to evaluate the fermionic determinant in Eq. (3.23). To this end, one proceeds exactly as in the Abelian case using the “decoupling technique.” Details of this calculation in the light-cone gauge for the $A_{\text{cl}}^{(m)} = 0$ case have been given in Ref. 10. In this particular case the answer is

$$\det D[a[h]] = \exp(-S_{\text{WZ}}[h]) \det(i\partial), \quad (3.24)$$

where S_{WZ} is the Wess-Zumino-Witten^{13,25} action:

$$\begin{aligned} S_{\text{WZ}}[h] = \frac{1}{2\pi} \text{tr} \int d^2x \partial_\mu h^{-1} \partial_\mu h \\ + \frac{1}{4\pi} e^{ijk} \text{tr} \int_B d^3y h^{-1} \partial_i h h^{-1} \partial_j h h^{-1} \partial_k h \end{aligned} \quad (3.25)$$

with B a ball with two-dimensional space-time as a

boundary.

For the $A_{\text{cl}}^{(m)} \neq 0$ case, one gets, instead of Eq. (3.24),

$$\det D[A_{\text{cl}}^{(m)} - a[h]] = \exp(-S_{\text{WZ}}[A_{\text{cl}}^{(m)}, h]) \det D[A_{\text{cl}}^{(m)}], \quad (3.26)$$

where $S_{\text{WZ}}[A_{\text{cl}}^{(m)}, h]$ is the gauged Wess-Zumino action

$$S_{\text{WZ}}[A_{\text{cl}}^{(m)}, h] = S_{\text{WZ}}[h] - S_C[A_{\text{cl}}^{(m)}, h] \quad (3.27)$$

with $S_C[A_{\text{cl}}^{(m)}, h]$ given by

$$\begin{aligned} S_C[A, h] = \frac{e^2}{4\pi} \text{tr} \int d^2x \left[A_+ h A_- h^{-1} + A_+ A_- \right. \\ \left. - \frac{i}{e} h^{-1} \partial_+ h A_- \right. \\ \left. + \frac{i}{e} h^{-1} \partial_- h A_+ \right]. \end{aligned} \quad (3.28)$$

We then finally have, for the QCD₂ generating functional, the bosonized form

$$Z = \sum_m \det \mathcal{D} [A_{\text{cl}}^{(m)}] \int Dh \exp(-S_{\text{eff}}^{\text{bos}} [A_{\text{cl}}^{(m)}, h]) \quad (3.29)$$

with

$$S_{\text{eff}}^{\text{bos}} [A_{\text{cl}}^{(m)}, h] = (2N+1)S_{\text{WZ}} [A_{\text{cl}}^{(m)}, h] + S_A [A_{\text{cl}}^{(m)} - a[h]] . \quad (3.30)$$

Here $S_{\text{WZ}} [A_{\text{cl}}^{(m)}, h]$ is given by Eq. (3.27) and $S_A [A_{\text{cl}}^{(m)} - a[h]]$ is the Yang-Mills action (3.4) written in terms of $A_{\text{cl}}^{(m)}$ and $a[h]$ through Eqs. (3.11)–(3.14).

Expression (3.29) is the non-Abelian extension of Eqs. (2.16) and (2.17) for the Schwinger model. The effective action $S_{\text{eff}}^{\text{bos}}$ gives the bosonized action for a non-Abelian fermionic model when topologically nontrivial sectors are taken into account. We shall discuss its properties in short. Concerning the fermion determinant appearing in Eq. (3.29), the Dirac operator has, as in the Abelian case, zero modes which have been studied in Refs. 26 and 27. The outcome is that, for topological charge $n > 0$ ($n < 0$), there are Nn ($N|n|$) square-integrable zero modes η_L (η_R) analogous to those found in the Abelian case. Indeed, one has

$$\eta_R^{(m,i)j} = \begin{bmatrix} z^m h_{ij}(z, \bar{z}) \\ 0 \end{bmatrix} , \quad (3.31)$$

$$\eta_L^{(m,i)j} = \begin{bmatrix} 0 \\ \bar{z}^{-m} h_{ij}^{-1}(z, \bar{z}) \end{bmatrix} , \quad (3.32)$$

with

$$h_{ij}(z, \bar{z}) = \exp[h(z, \bar{z})M_{ij}] \quad (3.33)$$

and

$$M = \frac{1}{N} \text{diag}(1, 1, \dots, 1-N) . \quad (3.34)$$

Here, $i, j = 1, 2, \dots, N$ and $m = 0, 1, \dots, |n| - 1$. In Eqs. (3.31) and (3.32), (m, i) labels the Nn ($N|n|$) different zero modes while a superscript j indicates the color fermion index. Concerning $h(z, \bar{z})$ in (3.33), it is defined as in the Abelian case by Eq. (2.20) once one uses, for $A_{\mu\text{cl}}^{(m)}$ in (3.9), the solution presented in Refs. 21 and 22.

We thus see that the partition function for QCD₂ only picks a contribution from the $n = 0$ sector due to the vanishing of the fermion determinant in Eq. (3.29) for any $n \neq 0$. However, as happens in the Abelian case, certain correlation functions, which vanish if one only takes into account the $n = 0$ sector, become nontrivial when $n \neq 0$ sectors are incorporated. One again needs a number of Fermi field factors such that the fermionic zero-mode contribution gets absorbed. A *minimal* (simplest nonvanishing) correlation function satisfying this criterion is, for the m th topological sector,

$$\left\langle \prod_{i=1}^N \prod_{\alpha=1}^{|m|} s_{\pm}(x_i^{\alpha}) \right\rangle = F_{\pm}(x_1^1, x_1^2, \dots, x_N^{|m|}) , \quad (3.35)$$

with

$$s_+(x) = \bar{\psi}_R^i(x) \psi_R^i(x) , \quad (3.36)$$

$$s_-(x) = \bar{\psi}_L^i(x) \psi_L^i(x) . \quad (3.37)$$

As in the Abelian case, the free fermionic part factorizes from the bosonic one:

$$F_{\pm}(x_1^1, x_1^2, \dots, x_N^{|m|}) = F_{\pm\text{fer}}(x_1^1, x_1^2, \dots, x_N^{|m|}) F_{\pm\text{bos}}(x_1^1, x_1^2, \dots, x_N^{|m|}) . \quad (3.38)$$

The fermionic factor $F_{\pm\text{fer}}(x_1^1, x_1^2, \dots, x_N^{|m|})$ can be computed exactly. The answer is

$$F_{+\text{fer}}(x_1^1, x_1^2, \dots, x_N^{|m|}) = (-1)^s \prod_{i,\alpha} V(z_i^{\alpha}) V(\bar{z}_i^{\alpha}) \prod \exp \left[2 \sum_i h(z_i^{\alpha}, \bar{z}_i^{\alpha}) M_{\alpha\alpha} \right] , \quad (3.39)$$

with

$$s = \frac{m(m+1)N}{2} + \frac{N(N+1)m}{2} , \quad (3.40)$$

$$V(z_i^{\alpha}) = \sum_{\{\sigma\}} (-1)^{\sigma} (z_i^{\alpha})^{\sigma_i} , \quad (3.41)$$

and $\{\sigma\} = \{\sigma_1, \sigma_2, \dots, \sigma_m\}$ representing a permutation of $\{1, 2, \dots, m\}$ with parity σ . (One gets an analogous expression for $F_{-\text{fer}}$.)

Concerning the bosonic factor, the non-Abelian character of the model complicates the calculations. One can envisage, however,¹⁰ an expansion of the form

$$h = \exp(2i\phi) = 1 + 2i\phi + \dots \quad (3.42)$$

in the effective action (3.30). One gets

$$S_{\text{eff}}^{\text{bos}} [A_{\text{cl}}^{(m)}, h[\phi]] = \frac{1}{2} \text{tr} \int d^2x (F_{\mu\nu}^{\text{cl}})^2 + S_0[\phi] + S_{\text{int}}[A^{\text{cl}}, \phi] . \quad (3.43)$$

As explained in Ref. 10, the action S_0 ,

$$S_0 = \frac{1}{4g^2} \left[\phi^a \left[\nabla \nabla - \frac{g^2}{\pi} \nabla \right] \phi^a \right] , \quad (3.44)$$

describes the dynamics of $N^2 - 1$ massive scalars (together with the $N^2 - 1$ nonphysical massless gauge excitations first discussed by Lowenstein and Swieca in their study of the Schwinger model²⁸). Concerning S_{int} , it takes a rather complicated looking form which will not be displayed here. One can however envisage the analysis of simple correlation functions along the lines of Refs. 29–31. In any case, the actual form of $F_{\pm\text{bos}}$ can be calculated using S_0 as a free (quadratic) action and taking all other terms arising from (3.43) as interactions. We shall not pursue this calculation in this paper. What we want to stress here is that bosonization using the path-integral approach allows us to handle the problem posed by nontrivial topological sectors, giving a systematic procedure to compute correlation functions which would vanish if only the trivial sector is taken into account.

IV. SUMMARY AND DISCUSSION

Following up the study of path-integral bosonization of two-dimensional fermionic models in topologically non-trivial sectors, initiated in Refs. 17 and 18 for the Thirring and Schwinger models, we considered in this paper non-Abelian models.

Using a non-Abelian vortex as a topological background, we have been able to derive the effective bosonic action associated with QCD₂, Eqs. (3.29) and (3.30). The corresponding partition function receives a contribution solely from the topologically trivial ($n=0$) sector but, as happens in the Abelian case,^{17,18,16} we have shown that certain fermionic correlation functions become nonvanishing provided one takes into account $n \neq 0$ sectors. Both results are due to the existence of zero modes for the Dirac operator in a topologically nontrivial background: the fermion determinant vanishes for $n \neq 0$ and, hence, only $n=0$ contributes to Z ; however, if one considers products of chiral-charge-nonconserving operators [of the type of Eqs. (3.36) and (3.37)], then the zero-mode contribution is absorbed (provided one has the appropriate number operator factors according to the number of zero modes).

Although we have considered QCD₂ with one flavor, the multiflavor case can be identically treated: the result for nontrivial correlation functions in $n \neq 0$ sectors will be just a product of identical one-flavor factors, one for each flavor species. One can also treat the case of purely fermionic models such as the SU(N) Thirring model or the chiral Gross-Neveu model, extending the approach presented in Ref. 6 for the $n=0$ sector. Indeed, one can eliminate the quartic fermionic self-interaction S_{int} characterizing these models,

$$S_{\text{int}} = -\frac{1}{2g^2} \int d^2x (\bar{\psi} \gamma_\mu t^a \psi)^2, \quad (4.1)$$

by introducing an auxiliary vector field A_μ taking values in the Lie algebra of SU(N) [U(N)], for the non-Abelian Thirring model (chiral Gross-Neveu model). [Accordingly, t^a are the generators of SU(N) or U(N).] Then, one

uses the identity

$$\exp(-S_{\text{int}}) = \int D A_\mu \exp \left[- \int d^2x (\bar{\psi} A \psi + \frac{1}{2} A_\mu^2) \right] \quad (4.2)$$

and proceeds as for the QCD₂ case. We expect that, as happens in the U(1) Thirring model, the inclusion of topologically non-trivial sectors has effects only in the $g \rightarrow \infty$ limit.¹⁷

We end this paper by noting that interest in handling topologically nontrivial sectors within the path-integral approach has recently revived due to the observation that fermion-number violation at high temperatures or at high energies may become important due to instanton effects.³²⁻³⁵

Indeed, although amplitudes leading to baryon- and lepton-number violations are in principle suppressed (as originally observed by 't Hooft³⁶) by an exponentially small Wentzel-Kramers-Brillouin (WKB) factor $\exp(-16\pi^2/g^2)$ with g the gauge coupling constant, inclusive rates of B - and L -violating processes occur frequently at high temperatures so as to considerably enhance these amplitudes. A similar behavior should be expected at high energies although there is, at present, some controversy concerning the way instanton calculations should be adequately adapted to analyze high-energy processes.^{37,38} The study of two-dimensional models such as the Abelian Higgs model with massless fermions^{29,30,38} can be of help in understanding what is going on in more complicated models. The approach we presented here, together with an instanton calculation along the lines of Ref. 31, may then shed light in the analysis of high-energy processes. We hope to report on this issue in a separate work.³⁹

ACKNOWLEDGMENTS

F.A.S. wishes to thank the Laboratoire de Physique Théorique et Hautes Energies (LPTHE) for hospitality and Fundación Antorchas (Argentina) and Centre National della Recherche Scientifique (CNRS) (France) for financial support. LPTHE is Laboratoire associé No. UA280 au CNRS.

-
- ¹R. Roskies and F. A. Schaposnik, Phys. Rev. D **23**, 558 (1981).
²R. Gamboa Saravi, F. A. Schaposnik, and J. Solomin, Nucl. Phys. **B185**, 239 (1981).
³R. Gamboa Saravi, F. A. Schaposnik, and J. Solomin, Phys. Rev. D **30**, 1353 (1984).
⁴S. Coleman, Phys. Rev. D **11**, 2088 (1975).
⁵S. Mandelstam, Phys. Rev. D **11**, 3026 (1975).
⁶K. Furuya, R. E. Gamboa Saravi, and F. A. Schaposnik, Nucl. Phys. **B208**, 159 (1982).
⁷E. Moreno and F. A. Schaposnik, Int. J. Mod. Phys. A **4**, 2827 (1989).
⁸F. A. Schaposnik, in *Particles and Fields*, Proceedings of the IV Swieca School, edited by A. da Silva *et al.* (World Scientific, Singapore, 1987), and references therein.
⁹R. Roskies, in *Symmetries in Particle Physics*, edited by I. Bars

- et al.* (Plenum, New York, 1984).
¹⁰E. Fradkin, C. M. Naón, and F. A. Schaposnik, Phys. Rev. D **36**, 3809 (1987).
¹¹K. Fujikawa, Phys. Rev. Lett. **42**, 1195 (1979); Phys. Rev. D **21**, 2848 (1980).
¹²J. Schwinger, Phys. Rev. **125**, 397 (1962).
¹³J. Wess and B. Zumino, Phys. Lett. **37B**, 95 (1971).
¹⁴C. M. Naon, Phys. Rev. D **31**, 2035 (1986).
¹⁵R. Gamboa Saravi, M. A. Muschietti, F. A. Schaposnik, and J. Solomin, Ann. Phys. (N.Y.) **15**, 360 (1984).
¹⁶K. Bardakci and L. Crescimanno, Nucl. Phys. **B313**, 269 (1989).
¹⁷M. V. Manias, C. M. Naon, and M. Trobo, Int. J. Mod. Phys. A **5**, 2853 (1990).
¹⁸M. V. Manias, C. M. Naon, and M. Trobo, Phys. Rev. D **41**,

- 3174 (1990).
- ¹⁹R. Jackiw and P. Rossi, Nucl. Phys. **B190**, 681 (1981).
- ²⁰H. B. Nielsen and P. Olesen, Nucl. Phys. **B61**, 45 (1973).
- ²¹H. J. de Vega and F. A. Schaposnik, Phys. Rev. D **14**, 1100 (1976).
- ²²H. J. de Vega and F. A. Schaposnik, Phys. Rev. D **34**, 3206 (1986).
- ²³G. Lozano, M. V. Manias, and F. A. Schaposnik, Phys. Rev. D **38**, 601 (1988).
- ²⁴A. M. Polyakov, in *Fields, Strings, and Critical Phenomena*, Proceedings of the Les Houches Summer School, Les Houches, France, 1988, edited by E. Brezin and J. Zinn-Justin, Les Houches Summer School Proceedings Vol. 49 (North-Holland, Amsterdam, 1990).
- ²⁵E. Witten, Commun. Math. Phys. **92**, 455 (1984).
- ²⁶H. J. de Vega, Phys. Rev. D **18**, 2932 (1978).
- ²⁷L. F. Cugliandolo and G. Lozano, Phys. Rev. D **39**, 3093 (1989).
- ²⁸J. Lowenstein and J. A. Swieca, Ann. Phys. (N.Y.) **68**, 172 (1971).
- ²⁹L. McLerran, A. Vainshtein, and M. Voloshin, Phys. Rev. D **42**, 171 (1990).
- ³⁰J. Kripfganz and A. Ringwald, Mod. Phys. Lett. A **5**, 675 (1990).
- ³¹F. A. Schaposnik, Phys. Rev. D **18**, 1183 (1978).
- ³²V. A. Kuzmin, V. A. Rubakov, and M. E. Shaposhnikov, Phys. Lett. **155B**, 36 (1985).
- ³³P. Arnold and L. McLerran, Phys. Rev. D **36**, 581 (1987).
- ³⁴A. Ringwald, Nucl. Phys. **B330**, 1 (1990).
- ³⁵O. Espinosa, Nucl. Phys. **B343**, 310 (1990).
- ³⁶G. 't Hooft, Phys. Rev. Lett. **37**, 8 (1976).
- ³⁷R. Singleton, L. Susskind, and L. Thorlacius, Nucl. Phys. **B343**, 541 (1990).
- ³⁸T. Banks, G. Farrar, M. Dine, D. Karabali, and B. Sakita, Rutgers University Report No. RU 90-20 (unpublished).
- ³⁹M. C. von Reichenbach, F. A. Schaposnik, and M. Trobo (unpublished).